

CHE 210 HW#2

1. A chemical reaction $A \rightarrow B$ (closed vessel)

t(min)	0	36	65	100	160	∞
$C_A(g/L)$	0.1823	0.1453	0.1216	0.1025	0.0795	0.0495

$$\ln \left(\frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} \right) = -kt + 0$$

$$C_{A0} = 0.1823$$

$$K = -0.0093$$

$$C_{A\infty} = 0.0495$$

$$R^2 = 0.9998$$

✓ 20/20

a. The data supports this prediction b/c the R^2 value is fairly close to 1.

b. 30.5 gal

$$K = \text{min}^{-1}$$

at $t=120$,

$$\ln \left(\frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} \right) = -kE$$

$$C_A =$$

$$C_{A0} =$$

$$C_{A\infty} =$$

$$t = 120 \text{ min}$$

$$C_{A0} \quad C_{A\infty, 2 \text{ hrs}}$$

$$21.04 \rightarrow 10.74 \in 10.30 \text{ B}$$

$$21.04 - X \rightarrow 10.74 - X$$

$$\ln X = a \leftrightarrow e^a = X$$

$$\log_{10}(X) = a$$

$$10^a = X$$

$$\ln X = a$$

$$\log_e(X) = a$$

As a decreases B, increases replacing

A spot

$$\lg 11 \quad 3.78541$$

$$30.5 \text{ gal} \rightarrow L = 115.455$$

$$\frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} = e^{-kE} (C_{A0} - C_{A\infty})$$

$$C_{A0} = 0.1823$$

$$C_{A\infty} = 0.093036237$$

$$C_{A0} - C_{A\infty} = 0.089263763$$

$$V \text{ conc} = m/v$$

$$C_{A0} \cdot V = m$$

$$m_A = 21.04$$

$$C_A = e^{-kt} (C_{A0} - C_{A\infty}) + C_{A\infty}$$

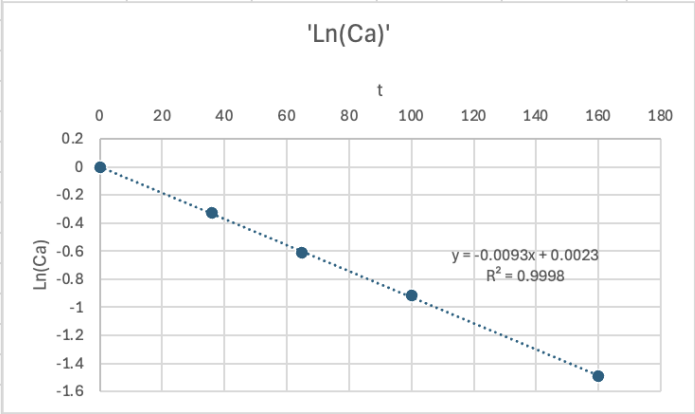
$$g/L \cdot V_L \quad \frac{M_A}{M_B} =$$

initial

$$0.510679 \text{ g}$$

$$m_B = 10.74$$

t	Ln(Ca)	Ca							
0	0	0.1823				Ca inf	0.0495		
36	-0.3265816	0.1453				Ca_0	0.1823		
65	-0.6107902	0.1216							
100	-0.9185523	0.1025							
160	-1.4876469	0.0795							



k 0.0093x

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2a. $r = \frac{Az}{B+z}$

Linearization 1 $\frac{1}{r} = \frac{B}{A} \cdot \frac{1}{z} + \frac{1}{A}$

$x_1 = 1/z$, $y_1 = 1/r$ $y_1 = m_1 x_1 + b_1$

$m_1 = \frac{B}{A}$ | $b_1 = \frac{1}{A}$ | $A = \frac{1}{b_1}$ | $B = \frac{m_1}{b_1}$

Linearization 2 $\frac{z}{r} = \frac{1}{A}(z) + \frac{B}{A}$

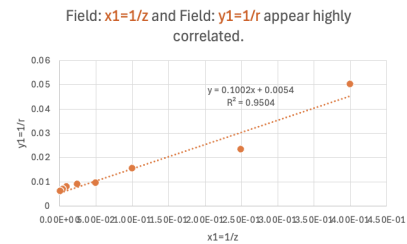
$x_2 = z$, $y_2 = z/r$ $y_2 = m_2 x_2 + b_2$

$m_2 = \frac{1}{A}$ | $b_2 = \frac{B}{A}$ | $A = \frac{1}{m_2}$ | $B = b_2 \cdot A$

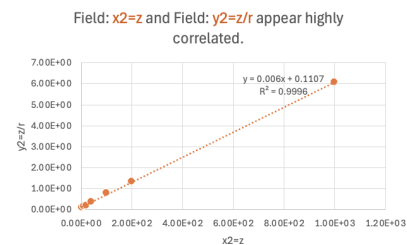
25/25

z	r
2.50E+00	20
4.00E+00	43
1.00E+01	65
2.00E+01	106
4.00E+01	112
1.00E+02	128
2.00E+02	150
1.00E+03	165

Linearization 1	
x1=1/z	y1=1/r
4.00E-01	0.05
2.50E-01	0.0232558
1.00E-01	0.0153846
5.00E-02	0.009434
2.50E-02	0.0089286
1.00E-02	0.0078125
5.00E-03	0.0066667
1.00E-03	0.0060606
y=0.1002+0.0054	
m1	0.1002
b1	0.0054
A=1/b1	185.18519
B=m1/b1	18.555556



Linearization 2	
x2=z	y2=z/r
2.50E+00	1.25E-01
4.00E+00	9.30E-02
1.00E+01	1.54E-01
2.00E+01	1.89E-01
4.00E+01	3.57E-01
1.00E+02	7.81E-01
2.00E+02	1.33E+00
1.00E+03	6.06E+00
y=0.006x+0.1107	
m2	0.006
b2	0.1107
A=1/m2	158.50887
B=b2*A	13.317494



r=Az/(B+z)		1/z	1/r	z/r	r_0	r_0-r	<-- squared
z	r						
2.5	20	4.00E-01	0.05	0.125	25.052778	5.0527783	25.530569
4	43	2.50E-01	0.0232558	0.0930233	36.612426	-6.387574	40.801096
10	65	1.00E-01	0.0153846	0.1538462	67.978519	2.9785185	8.8715727
20	106	5.00E-02	0.009434	0.1886792	95.150536	-10.84946	117.71087
40	112	2.50E-02	0.0089286	0.3571429	118.91697	6.9169706	47.844482
100	128	1.00E-02	0.0078125	0.78125	139.88032	11.880319	141.14199
200	150	5.00E-03	0.0066667	1.3333333	148.6131	-1.386898	1.9234854
1000	165	1.00E-03	0.0060606	6.0606061	156.42567	-8.574325	73.519049

sum
457.34311
objective
min
changing A & B



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3. Murphy's P.1.12 pg. 43

- Price of all raw material stays the same, except glucose
- Using either benzene (C_6H_6), or glucose ($C_6H_{12}O_6$) as a raw material for making adipic acid $C_6H_{10}O_4$ or catechol ($C_6H_6O_2$)

How much would the price of glucose (\$/kg) have to drop for the glucose-to-adipic acid process to be economically competitive with the benzene-to-adipic acid process. What about the catechol process?

Adipic Acid Process		Atom Economy & Process Economy of Glucose-to-adipic acid process						
	Compound	V_i	M_i	$V_i M_i$	(SF=192.5 kg/d)	\$/kg	\$/day	
glucose	$C_6H_{12}O_6$	-7/3	180	-420	-80,850	0.60	-48,500	
	O_2	-8.5	32	-272	-52,360	0	0	
	CO_2	+8	44	+352	+67,760	0	0	
	H_2O	+11	18	+198	+38,120	0	0	
	H_2	-2	2	-4	-770	0.2	-150	
	$C_6H_{10}O_4$	+1		+146	+28,000	1.50	+43,300	
	Sum				0		-5,400	

pg. 36

pg. 36

Atom Economy & Process Economy for Benzene-to Adipic Acid Process							
Compound	V_i	M_i	$V_i M_i$	(SF=192.3 kg/d)	\$/kg	\$/day	
C_6H_6	-1	78	-78	-15,000	0.41	-6150	
H_2	-3	2	-6	-1,154	0.2	-230	
O_2	-1	32	-32	-6,154	0	0	
HNO_3	-2	63	-126	-24,230	0.40	-9700	
$C_6H_{10}O_4$	+1	146	+146	+28,076	1.54	+43,200	
NO	+2	30	+60	+11,538	0	0	
H_2O	+2	18	+36	+6,923	0	0	
Sum			0	0		+27,101	pg. 32

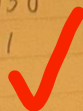
pg. 32

$$43,300 - 150 = 43,150$$

$$(-80,850)x + 43,150 = 27,100$$

$$(-80,850)x = -16,050$$

$$x = 0.19851$$



$0.60 - 0.19851 = 0.401484$ \$/kg
The price would have to drop approximately 0.4015 \$/kg to be economically competitive w/ benzene-to-adipic acid process

Catechol →
Process
glucose

Compound	V_i	Mg/g-mol	V_i, M_i	SF=192.34g/d	$\$/kg$	$\$/day$
$C_6H_{12}O_6$	-7/3	180	-420	-80,770	0.60	-48,500
O_2	-7.5	32	-240	-46,150	0	0
CO_2	+8	44	+352	+67,690	0	0
H_2O	+11	18	+198	+38,080	0	0
$o-C_6H_6O_2$	+1	110	+110	+21,150	4.62	+97,700
Sum				0		+49,200

Compound	V_i	M_i	V_i, M_i	SF=192.34g/d	$\$/kg$	$\$/day$
C_6H_6	-1	78	-78	-15,000	0.41	-6,150
C_3H_6	-1	42	-42	-8,077	0.26	-2,100
O_2	-1	32	-32	-6,154	0	0
H_2O_2	-1	34	-34	-6,538	1.49	-9,740
C_3H_6O	+1	58	+58	+11,154	0.86	+9,600
$o-C_6H_6O_2$	+1	110	+110	+21,153	4.62	+97,700
H_2O	+1	18	+18	+3,462	0	0
Sum				0		+89,300

Sum of other raw materials = 97,700

$$(-80,770)x + 97,700 = 89,300$$

$$(-80,770)x = -8400$$

$$x = 0.103999 \text{ \$/kg}$$

↖ $\$/kg$ glucose would have to be

$$0.60 - 0.103999 = 0.496001$$

The price would have to drop 0.496001 $\$/kg$ in order to be economically competitive with benzene-to-catechol-process

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4. Murphy P2.11

1970: emissions = 14 g/mile travel = 1100 billion miles

2000: emissions = 1 g/mile travel = 2600 billion miles

Convert (1 ton = 2000 lbm) (1 lbm = 453.592 g)

~~1970 total emissions = (14 g/mile) (1100 × 10⁹ miles) = 1.54 × 10¹³ g~~
~~grams → lbm 1.54 × 10¹³ g × $\frac{1 \text{ lbm}}{453.592 \text{ g}}$ = 3.40 × 10¹⁰ lbm~~

~~lbm → tons 3.40 × 10¹⁰ lbm / 2000 = 1.70 × 10⁷ tons = 17 million tons~~

~~2000 total emissions = (1 g/mile) (2600 × 10⁹ miles) = 2.6 × 10¹² g~~
~~grams → lbm 2.6 × 10¹² g × $\frac{1 \text{ lbm}}{453.592 \text{ g}}$ = 5.73 × 10⁹ lbm~~

-10 pts

~~5.73 × 10⁹ lbm / 2000 = 2.87 × 10⁶ tons~~
~~≈ 2.9 million tons~~

Percent change (1970 → 2000)

% change = $\frac{2.87 \times 10^6 - 1.70 \times 10^7}{1.70 \times 10^7} \times 100\% \approx -83\%$ ✓

There is a decrease in total hydrocarbon emissions over the 30 years.

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5. Mixture 60 wt% water $\rightarrow$ 40 wt% ethanol
 ρ (density) of water @ 20°C $\rho_w = 0.9982 \text{ g/mL}^{-1} = 998.1 \text{ g/L}^{-1}$
 ρ_e density of ethanol @ 20°C $\rho_e = 0.7893 \text{ g/mL}^{-1} = 789.3 \text{ g/L}^{-1}$
Molar Mass ethanol = 46.07 g/mol^{-1}

a. 100 g $\rightarrow$ 100 %

60 g. water $\rightarrow$ 60 %

40 g. ethanol $\rightarrow$ 40 %

$$V_w = \frac{m_w}{\rho_w} = \frac{60.0 \text{ g}}{998.1 \text{ g/L}^{-1}} = 0.060108194 \text{ L}$$

$$V_e = \frac{m_e}{\rho_e} = \frac{40.0 \text{ g}}{789.3 \text{ g/L}^{-1}} = 0.0506778158 \text{ L}$$

$$V_{\text{mix}} = V_w + V_e = 0.060108 + 0.0506778 \approx 0.11078601054 \text{ L}$$

$$\rho_{\text{mix}} = \frac{100 \text{ g}}{0.110786 \text{ L}} = 902.64 \text{ g/L}^{-1}$$

$$\text{Specific Gravity} = \frac{\rho_{\text{mix}}}{\rho_w} = \frac{902.64104209}{998.2} = 0.90427$$

Mass of ethanol needed (150 mol ethanol)

$$m_{e, \text{needed}} = 150 \times 46.07 = 6910.5 \text{ g}$$

(molar mass, e)

$$m_{\text{mix}} = \frac{m_{e, \text{needed}}}{0.40} = \frac{6910.5 \text{ g}}{0.40} = 17276.25 \text{ g}$$

$$V_{\text{needed}} = \frac{17276.25 \text{ g}}{902.6410429 \text{ g/L}^{-1}} = 19.139 \text{ L} \approx 19.14 \text{ L}$$

b. $\rho_{\text{mix}} = 1.50 \text{ given } (\rho_w) = 0.93518 (998.2 \text{ g/L}^{-1}) = 933.4966 \text{ g/L}^{-1}$

$$m_{\text{mix}} = 17276.25 \text{ g}$$

$$V_{\text{needed, act}} = \frac{17276.25 \text{ g}}{933.4966 \text{ g/L}^{-1}} = 18.5070 \text{ L} \approx 18.51 \text{ L}$$

$$\% \text{ Error} = \frac{V_{\text{add}} - V_{\text{act}}}{V_{\text{act}}} \times 100\% = \frac{19.13966 - 18.5070}{18.5070} \times 100\% \approx 3.41\%$$

overestimated
 $\downarrow$
 $\approx 3.42\%$